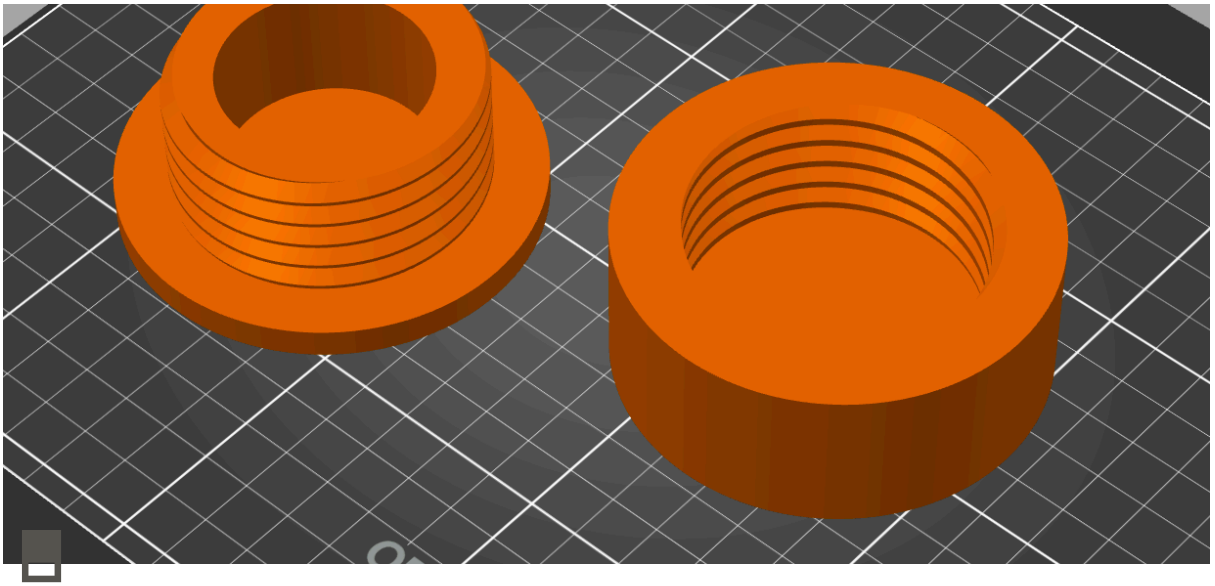


Cylinder

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Date	2025 autumn
School	Malmö University

Overview

This project explores a cylindrical 3D printed container designed to safely house a sensor in wet conditions. The main focus was making the object waterproof through threaded parts and tight tolerances, which is also visible in the key visuals. One of the images shows the container fully submerged, demonstrating that it can handle water without leaking. The form is simple and functional, with threads, fasteners, and print details clearly exposed, making the construction and behaviour easy to understand at a glance.

Reflection

The hardest part was definitely handling the threads. I didn't know at first how to implement them properly, and a lot of time was spent debugging. Setting up the print output alone took nearly 3 hours. I followed the tutorial for guidance, but it was still a big challenge. I learned that it's important to plan threads ahead, and next, I want to add a magnet feature so you can place and manipulate objects with it. I chose 3D printing over laser cut parts because it allowed the threads and seals to be part of the object itself, making the container sturdier and better suited for repeated use and exposure to water. I learned how material choice and thread design directly affect durability and usability, and I want to push functional details like sealing and attachment systems further.

Visuals



Figure 1. First trying.

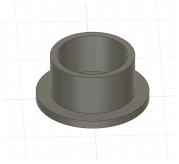


Figure 2. Second try, still not the final product. Cup.

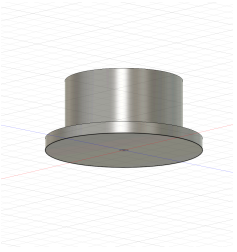


Figure 3. Figure 2 from another perspective (Downside). Cup.

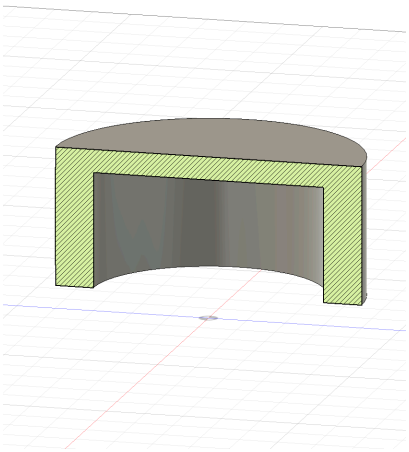


Figure 4. Lid

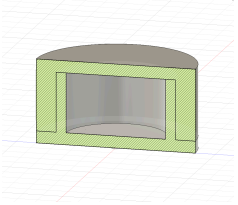


Figure 5. Lid and Cup together without threads.

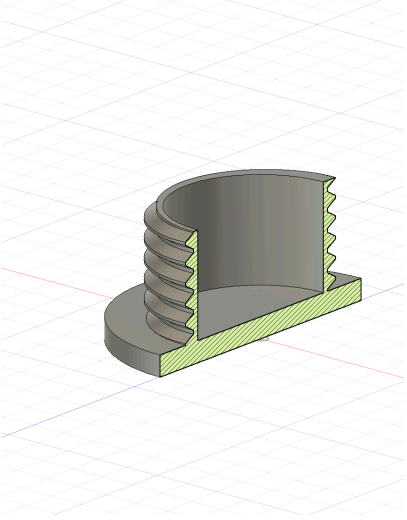


Figure 6. Creating threads 5.5x for the cup with full length and modelled.

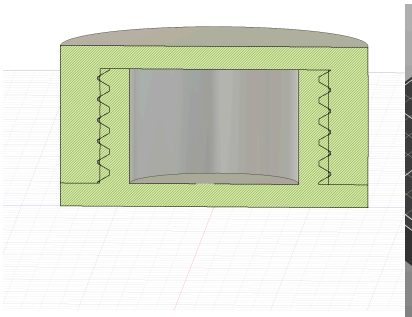


Figure 7. Some minor changes in size. But it is not worked. restarted it.

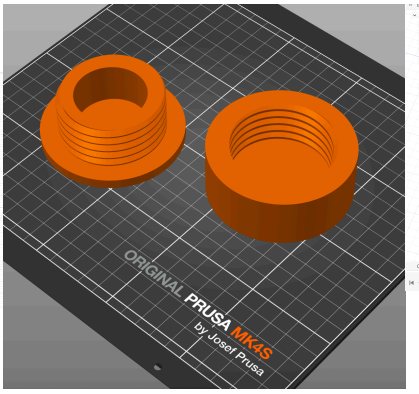


Figure 8. Not working threads.

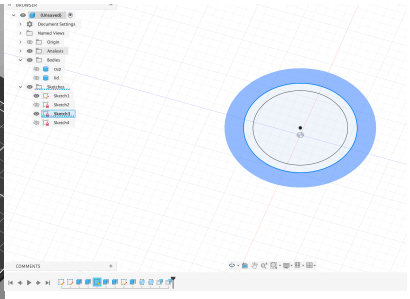


Figure 9. Restart sketches, fundamentals. Lat try, Working prototype.

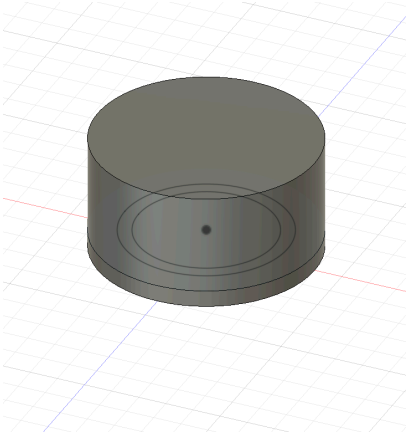


Figure 10. Sketching extruded and divided the Cup and the Lid from each other(Check Figure 12 for section analysis).

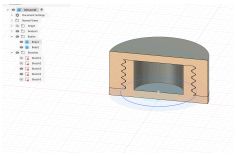


Figure 11. Threads, section analysis with sketches.

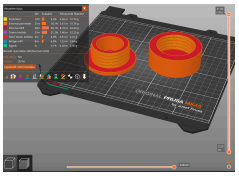


Figure 12. Prepared to Prusa Slicer for printing.

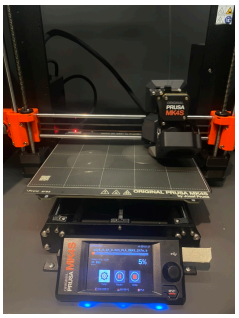


Figure 13. Started printing process.

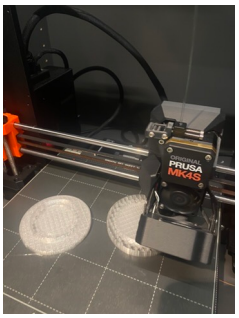


Figure 14. Printing Lid and Cup.

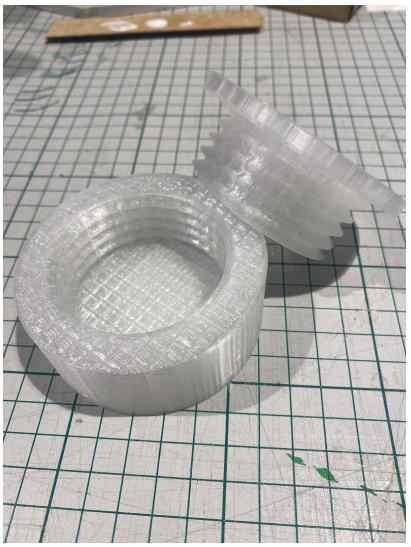


Figure 15. Final Prototype.



Figure 16. Testing under water (check video)

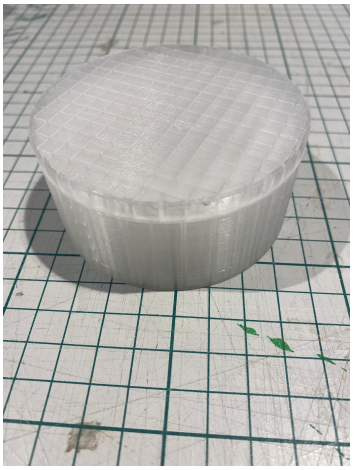


Figure 17. Closed Cylinder.

https://drive.google.com/file/d/1p9xvSOaQefCHLQRUG3SCETkZmLIb8IBy/view?usp=sharing

Reference

https://www.youtube.com/watch?v=eBte6lmHzOc